

AWS

Module-5

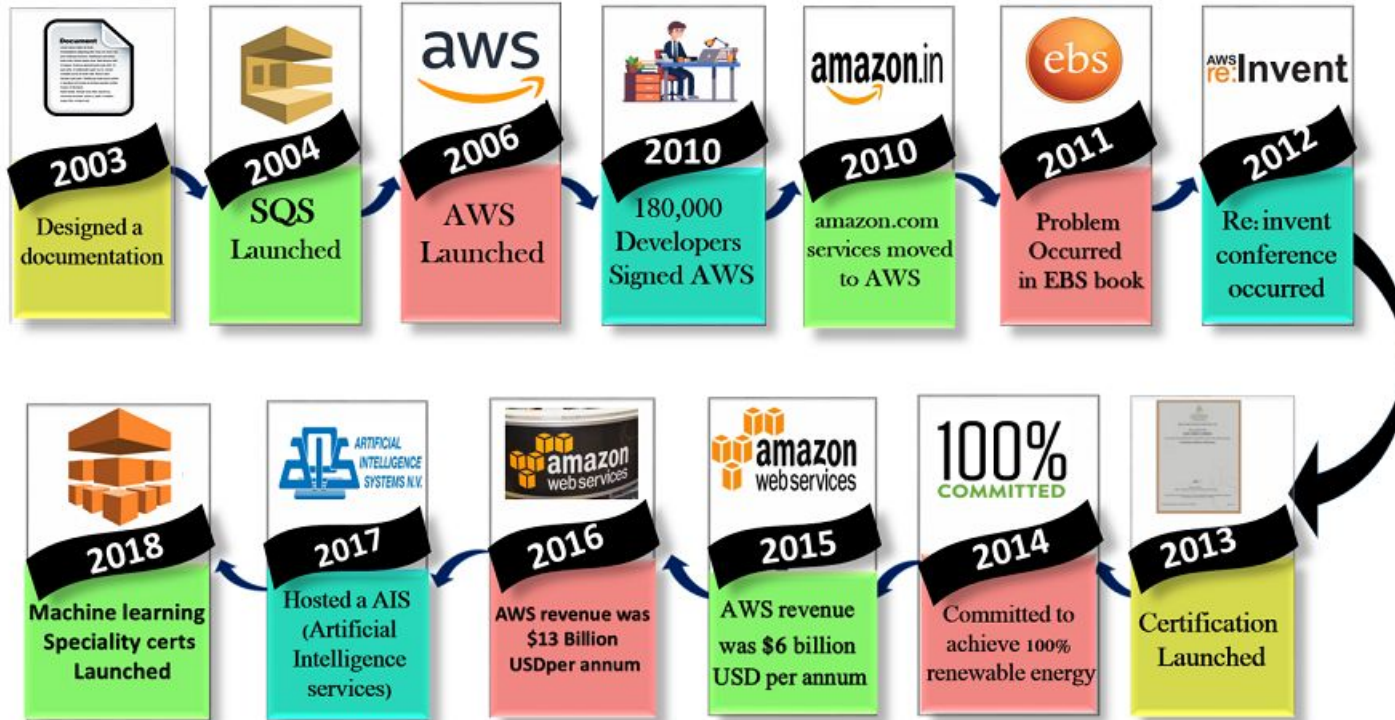
Cloud Implementations and Environments: Amazon Web Services, The Elastic Compute Cloud (EC2), The Simple Storage Service (S3), Elastic Block Store (EBS), The Simple Queuing Services (SQS), Identity and Access Management (IAM).

Google AppEngine - PaaS, Windows Azure, A Comparison of Cloud Computing Platforms.

What is AWS?

- AWS stands for Amazon Web Services.
- It is a collection of remote computing services (also called web services) that make up a cloud computing platform, offered by Amazon.com.
- As of mid-2025, AWS operates over 114 Availability Zones across approximately 36–37 geographic Regions worldwide, providing extensive global scale, high availability, and strong fault isolation.
- They offer a variety of services such as storage, networking, databases, analytics, application services, and deployment.
- AWS is one of the most popular cloud computing platforms and is widely used by individuals, companies and government organizations.

History of AWS?



What AWS Offers?

- Amazon Web Services (AWS) offers a wide range of services to help customers build, deploy, and manage their applications and infrastructure. These services are organized into several categories, including:
 - **Computing:** Services such as Amazon Elastic Compute Cloud (EC2), Amazon Elastic Container Service (ECS), and Amazon Elastic Kubernetes Service (EKS) provide customers with the ability to run their own virtual servers and containerized applications in the cloud.
 - **Storage:** Services such as Amazon Simple Storage Service (S3), Amazon Elastic Block Store (EBS), Amazon Elastic File System (EFS), and Amazon Glacier provide customers with scalable, highly available storage for their data.

What AWS Offers? (Cont'd)

- **Databases:** Services such as Amazon Relational Database Service (RDS), Amazon DynamoDB, Amazon DocumentDB, Amazon Neptune, and Amazon Redshift provide customers with the ability to run their own databases in the cloud.
- **Analytics:** Services such as Amazon QuickSight, Amazon EMR, Amazon Athena, Amazon Kinesis and Amazon Elasticsearch Service provide customers with the ability to collect, process, and analyze large sets of data.
- **Mobile:** Services such as Amazon Pinpoint, Amazon Mobile Analytics, and AWS AppSync provide customers with the ability to build, test, and monitor mobile apps.
- **IoT:** Services such as Amazon IoT, AWS IoT Greengrass, and AWS IoT Device Management provide

What AWS Offers? (Cont'd)

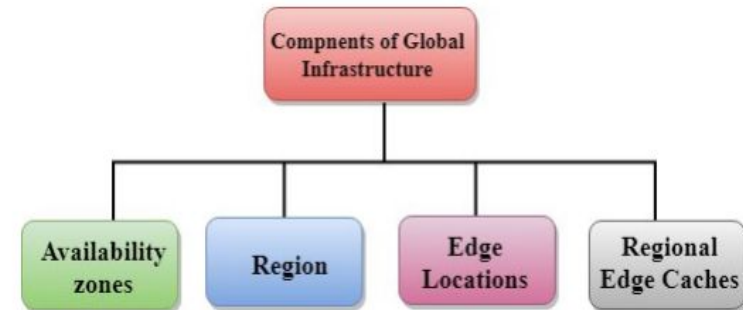
- **Application services:** Services such as Amazon Elasticsearch, Amazon Elastic Transcoder, Amazon AppSync, Amazon Pinpoint, and Amazon Connect provide customers with the ability to build, deploy, and manage various applications.
- **Deployment and Management:** Services such as AWS CloudFormation, AWS Elastic Beanstalk, AWS CodeDeploy and AWS CodePipeline provide customers with the ability to automate their infrastructure

AWS Global Infrastructure

- AWS Global Infrastructure refers to the network of data centers, availability zones, and edge locations that Amazon Web Services (AWS) uses to provide its services.
- As of August 2025, AWS spans 37 geographic regions and 117 Availability Zones.
- The following are the components that make up the AWS infrastructure:
 - Region
 - Availability Zone
 - Edge locations
 - Regional Edge Caches

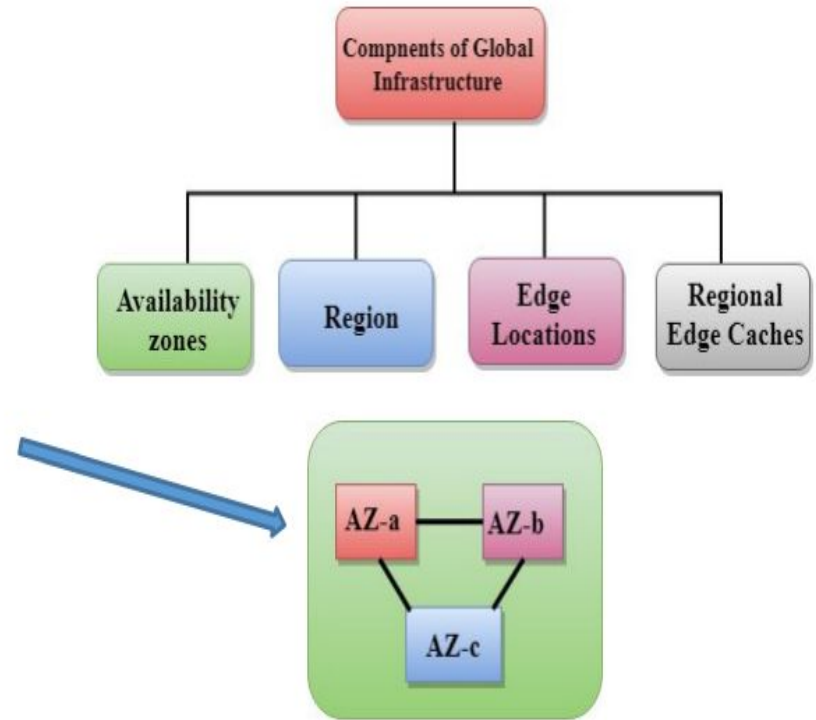
Availability zone as a Data Center

- An Availability Zone (AZ) is one or more discrete data center facilities located in a specific geographic area, typically within a single country or city.
- Each AZ contains multiple components such as servers, networking equipment (switches, routers), load balancers, and firewalls.
- All the infrastructure that processes and stores cloud data resides inside these data centers.
- A single Availability Zone can consist of multiple data centers; if they are geographically close and connected with low-latency links, they are treated as one AZ.



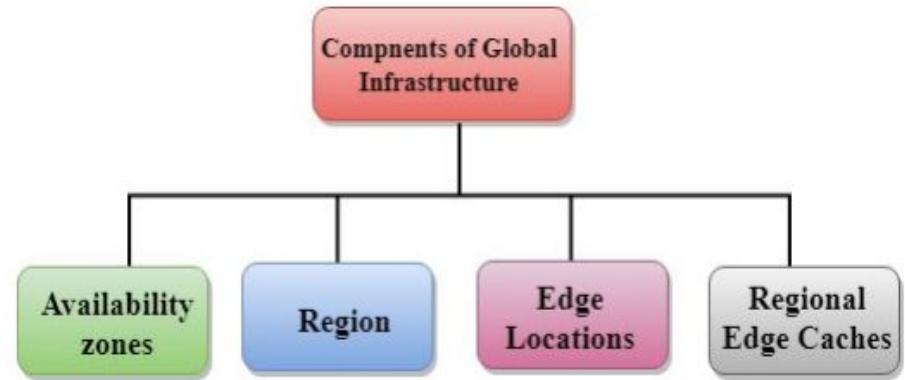
AWS Global Infrastructure Region

- A region is a geographical area.
- Each region consists of 2 more availability zones.
- A region is a collection of data centers which are completely isolated from other regions.
- A region consists of more than two availability zones connected to each other through links.
- Availability zones are connected through redundant and isolated metro fibers.



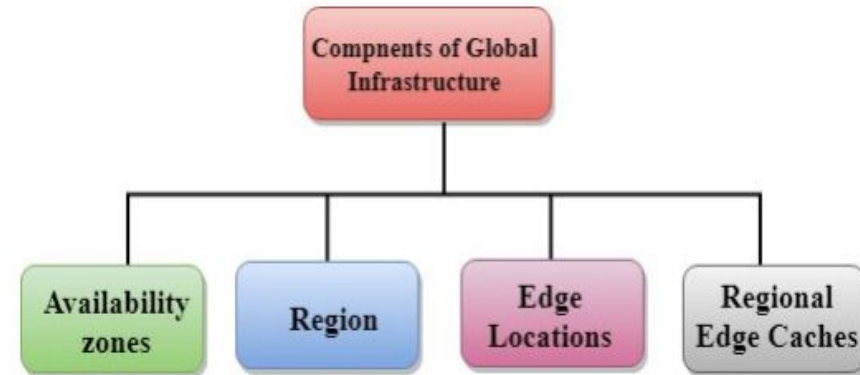
AWS Global Infrastructure Edge Locations

- Edge locations are the endpoints for AWS used for caching content.
- Edge locations consist of CloudFront, Amazon's Content Delivery Network (CDN).
- Edge location is not a region but a small location that AWS have.
- It is used for caching the content.
- Located in most of the major cities.
- For example, some user accesses your website from Singapore; then this request would be redirected to the edge location closest to Singapore where cached data can be read.

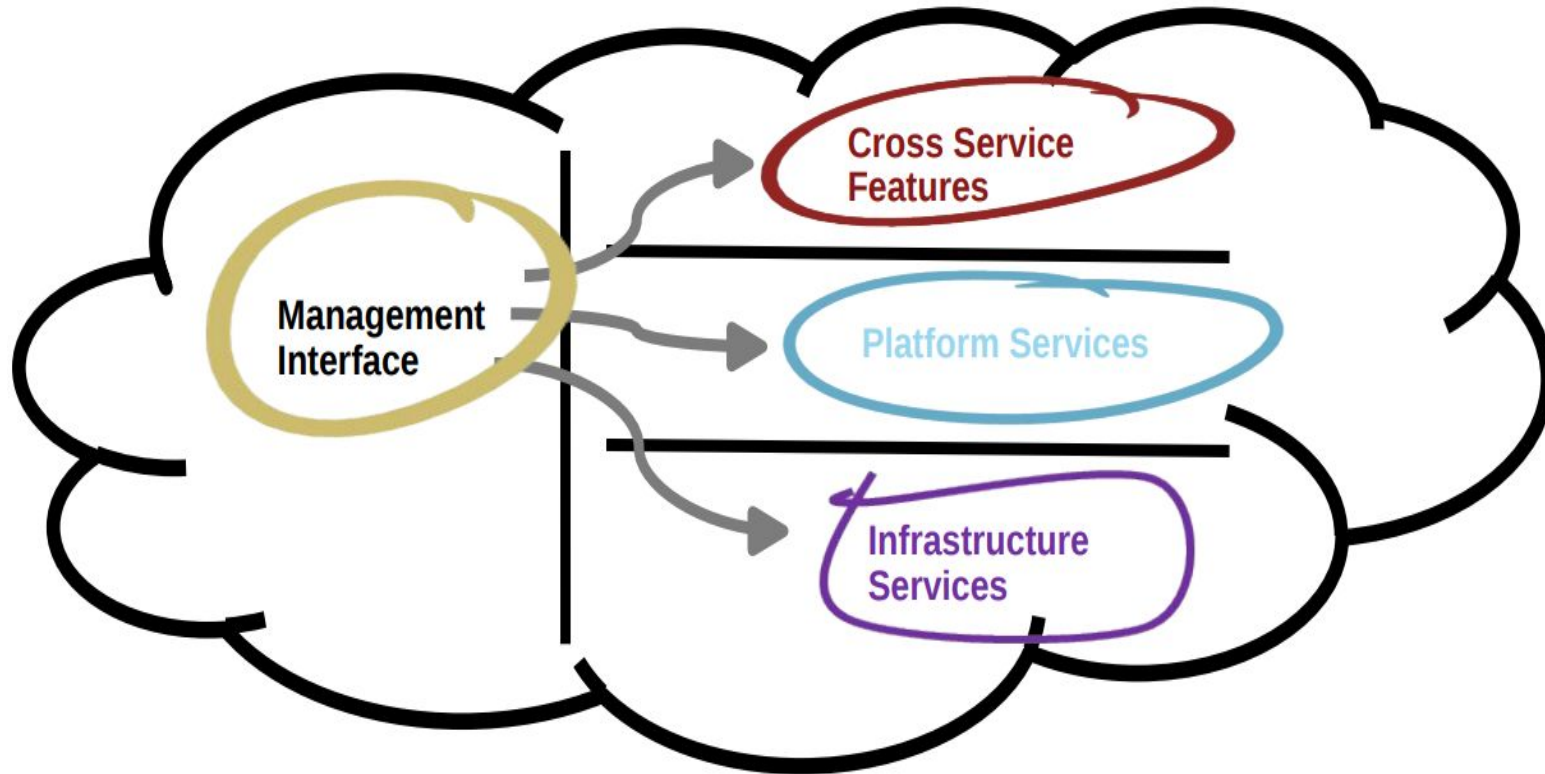


AWS Global Infrastructure Regional Edge Cache

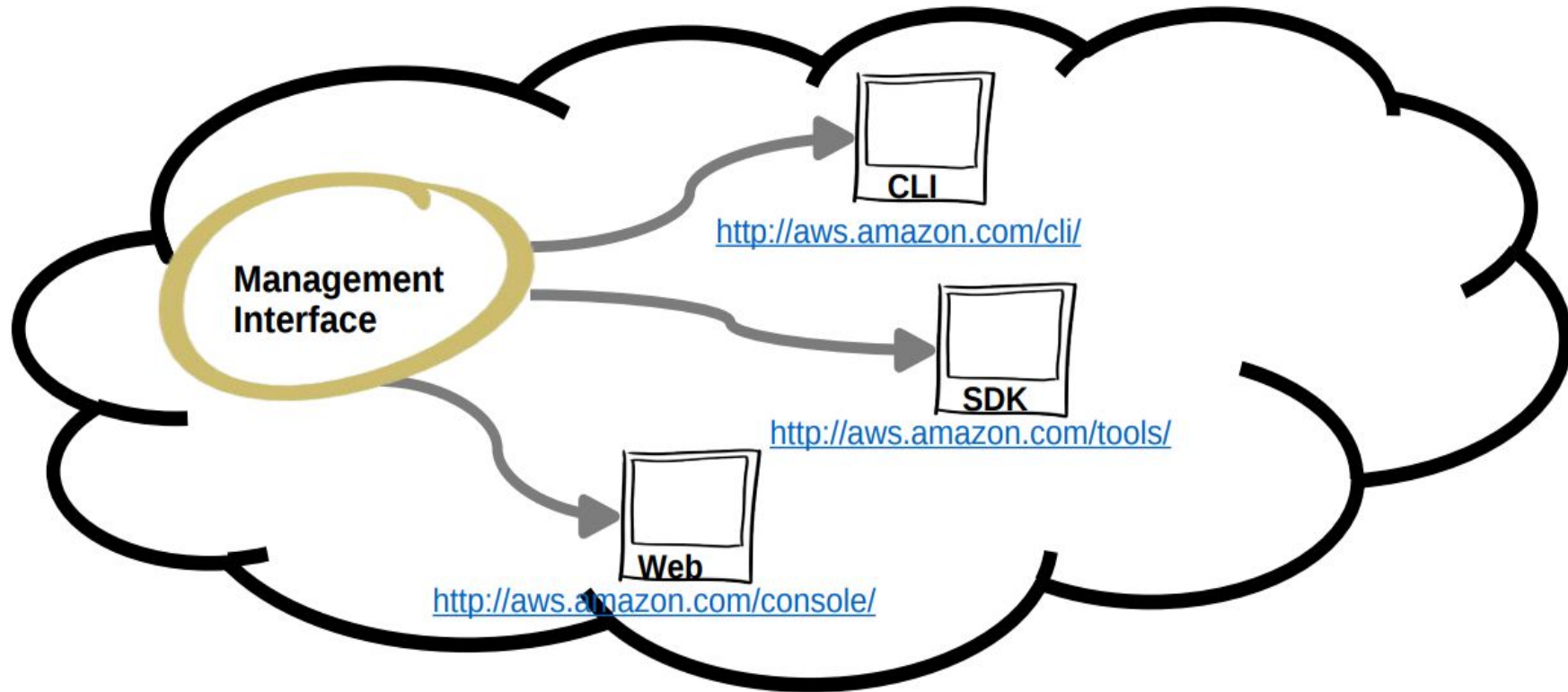
- AWS announced a new type of edge location in November 2016, known as a Regional Edge Cache.
- Regional Edge cache lies between Cloud-Front Origin servers and the edge locations.
- A regional edge cache has a large cache than an individual edge location.
- Data is removed from the cache at the edge location while the data is retained at the Regional Edge Caches.
- Edge location retrieves the cached data from the Regional edge cache instead of the Origin servers that have high latency.



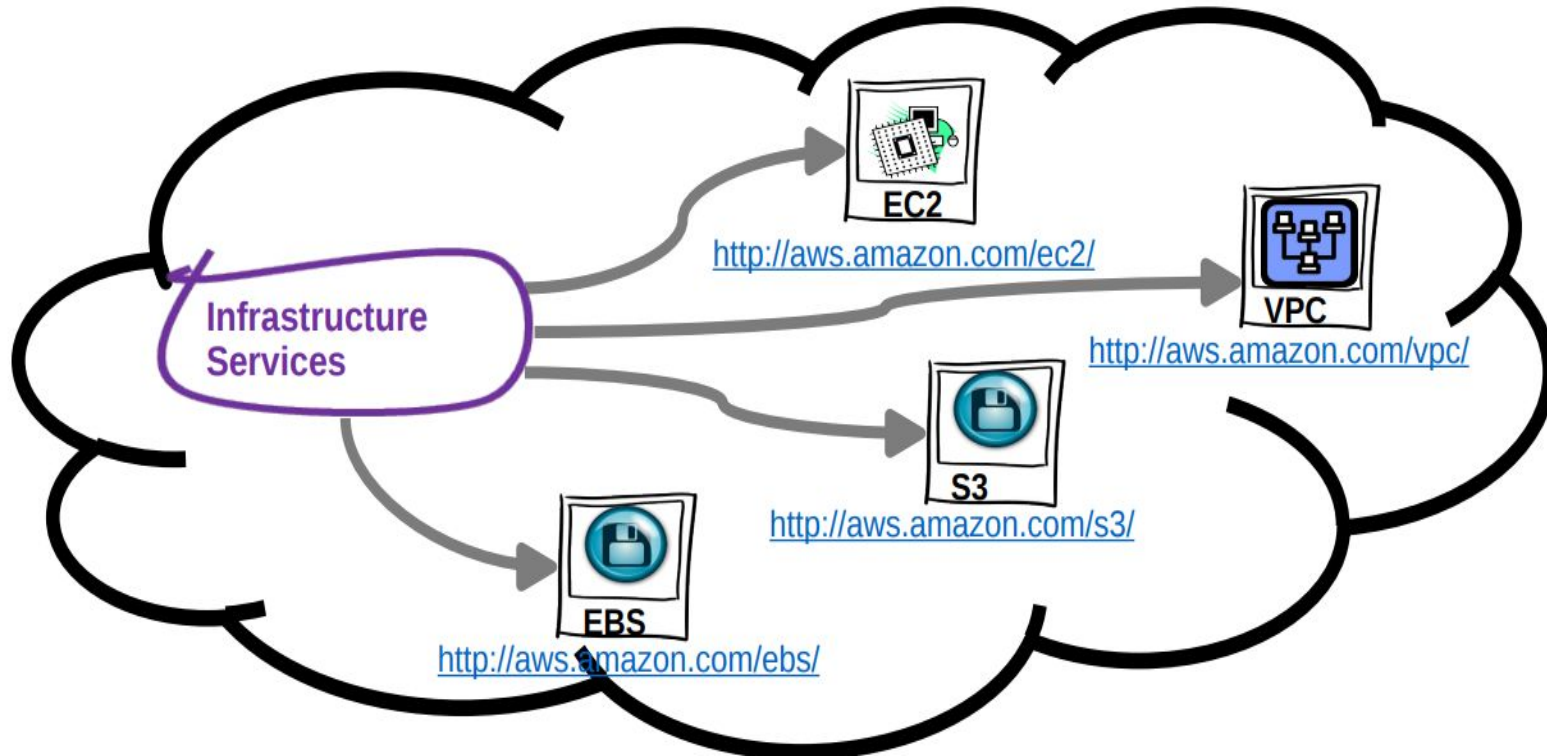
The Amazon Web Services Universe



Management Interface



Infrastructure Services



Compute Services

Amazon Elastic Compute Cloud (EC2)

Amazon Elastic MapReduce

AWS Elastic Beanstalk

AWS Cloudformation

Autoscaling

Storage Services

Amazon Simple Storage Service (S3)

Amazon Elastic Block Store (EBS)

Amazon ElastiCache

Amazon SimpleDB

Amazon Relational Database Service (RDS)

Amazon CloudFront

Amazon Import/Export

Communication Services

Amazon Simple Queue Service (SQS)

Amazon Simple Notification Service (SNS)

Amazon Simple Email Service (SES)

Amazon Route 53

Amazon Virtual Private Cloud (VPC)

Amazon Direct Connect

Amazon Elastic Load Balancing

Additional Services

Amazon GovCloud

Amazon CloudWatch

Amazon Flexible Payment Service (FPS)

Amazon DevPay

Amazon Fulfillment Web Service (AWS)

Amazon Mechanical Turk

Alexa Web Information Service

Alexa Top Sites

Compute Services



- AWS offer compute services from dozens of data centers spread across **availability zones (AZs)** in regions across the world.
- A business will choose **one or multiple** availability zones depending upon compliance and proximity to end customers.
- Amazon Elastic Compute Cloud (EC2) is a service that **provides virtual servers** called EC2 instances for compute capacity.
- The EC2 service offers **dozens of instance** types with **varying capacities** and sizes, tailored to specific workload types and applications, such as memory-intensive and accelerated-computing jobs.
- AWS also provides an **Auto Scaling tool** to dynamically scale capacity to maintain instance health and performance.

Amazon Elastic Compute Cloud (EC2)

- EC2 Stands for **Elastic Compute Cloud**.
- Provides **scalable computing** capacity in the Amazon Web Services (AWS) cloud.
- A web service that provides resizable compute capacity in the cloud.
- You can use Amazon EC2 to **launch as many or as few virtual servers** as you need, configure security and networking, and manage storage.
- Applications **anywhere** on the Internet can **launch a virtual server** in the amazon cloud with a single web services call.
- Amazon EC2 enables you to **scale up or down to handle** changes in requirements.

EC2 Features

- **Virtual computing** environments, known as *instances*
- **Preconfigured templates for your instances**, known as *Amazon Machine Images (AMIs)*, that package all requirements for your server.
- **Various configurations** of CPU, memory, storage, and networking capacity for your instances, known as *instance types*.
- **Secure login** information for your instances using *key pairs*.
- **Storage volumes** for temporary data that's deleted when you stop or terminate your instance, known as *instance store volumes*.
- **Persistent storage volumes** for your data using Amazon Elastic Block Store (Amazon EBS), known as *Amazon EBS volumes*.

EC2 Features

Multiple physical locations for your resources, such as instances and Amazon EBS volumes, known as *Regions* and *Availability Zones*.

A **firewall** that enables you to specify the protocols, ports, and source IP ranges that can reach your instances using *security groups*.

Static IPv4 addresses for dynamic cloud computing, known as *Elastic IP addresses*

Metadata, known as *tags*, that you can create and assign to your Amazon EC2 resources.

Virtual networks that can be optionally connected to your own network, known as *virtual private clouds* (VPCs).

Starting with EC2

- Launching an Amazon EC2 instance for the first time need the following steps:
 - Sign up for AWS
 - Create a key pair
 - Create a security group

Sign Up with EC2

To create an AWS account

Open <https://portal.aws.amazon.com/billing/signup>

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Follow the online instructions.

Create a Key Pair

- To create your key pair
- Open the **Amazon EC2 console** at <https://console.aws.amazon.com/ec2/>.
- In the navigation pane, choose **Key Pairs**.
- Choose Create key pair.
- For **Name**, enter a **descriptive name** for the key pair. Amazon EC2 associates the public key with the name that you specify as the key name. A key name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
- For **File format**, choose the format in which to save the private key.
- Choose Create key pair.

Create a Security Group

- **To create a security group with least privilege**
- Open the **Amazon EC2 console** at <https://console.aws.amazon.com/ec2/>.
- From the navigation bar, **select a Region for the security group**. Security groups are specific to a Region, so you should select the same Region in which you created your key pair.
- In the navigation pane, choose **Security Groups**.
- Choose **Create security group**.
- In the **Basic details** section, do the following:
 - Enter a name for the new security group and a description.
 - In the VPC list, select your default VPC for the Region.

Security Group

- In the **Inbound rules** section, create the following rules (choose **Add rule** for each new rule):
 - Choose **HTTP** from the **Type** list, and make sure that **Source** is set to **Anywhere** (0.0.0.0/0).
 - Choose **HTTPS** from the **Type** list, and make sure that **Source** is set to **Anywhere** (0.0.0.0/0).
 - Choose **SSH** from the **Type** list. In the **Source** box, choose **My IP** to automatically populate the field with the public IPv4 address of your local computer.
 - Choose **Create security group**.

- **EC2 allows creating Virtual Machines (VM)** on-demand. Pre-configured templated Amazon Machine Image (AMI) can be used get running immediately. Creating and sharing your own AMI is also possible via the AWS Marketplace.
- **Auto Scaling** allows automatically scale of the capacity up seamlessly during demand spikes to maintain performance, and scales down during demand lulls to minimize costs.
- **Elastic Load Balancing** automatically distributes incoming application traffic across multiple Amazon EC2 instances.
- Provide **tools to build failure resilient applications** by launching application instances in separate Availability Zones.
- Pay only for resources actually consume, instance-hours.
- **VM Import/Export** enables you to easily import virtual machine images from your existing environment to Amazon EC2 instances and export them back at any time.

EC2 Instances

Micro instances (t1.micro): – Micro Instance 613 MiB of memory, up to 2 ECUs (for short periodic bursts), EBS storage only, 32-bit or 64-bit platform.

- Standard Instances provide customers with a balanced set of resources and a low cost platform.
- M1 Small Instance (Default) 1.7 GiB of memory, 1 EC2 Compute Unit (1 virtual core with 1 EC2 Compute Unit), 160 GB of local instance storage, 32-bit or 64-bit platform
- M1 Medium Instance 3.75 GiB of memory, 2 EC2 Compute Units (1 virtual core with 2 EC2 Compute Units each), 410 GB of local instance storage, 32-bit or 64-bit platform
- M1 Large Instance 7.5 GiB of memory, 4 EC2 Compute Units (2 virtual cores with 2 EC2 Compute Units each), 850 GB of local instance storage, 64-bit platform
- M1 Extra Large Instance 15 GiB of memory, 8 EC2 Compute Units (4 virtual cores with 2 EC2 Compute Units each), 1690 GB of local instance storage, 64-bit platform
- M3 Extra Large Instance 15 GiB of memory, 13 EC2 Compute Units (4 virtual cores with 3.25 EC2 Compute Units each), EBS storage only, 64-bit platform
- M3 Double Extra Large Instance 30 GiB of memory, 26 EC2 Compute Units (8 virtual cores with 3.25 EC2 Compute Units each), EBS storage only, 64-bit platform

One EC2 Compute Unit (ECU) provides the equivalent CPU capacity of a 1.0-1.2 GHz 2007 Opteron or 2007 Xeon processor

EC2 High Performance Instances

- **High-Memory Instances:** – **High-Memory Extra Large Instance** 17.1 GiB memory, 6.5 ECU (2 virtual cores with 3.25 EC2 Compute Units each), 420 GB of local instance storage, 64-bit platform
- **High-Memory Double Extra Large Instance** 34.2 GiB of memory, 13 EC2 Compute Units (4 virtual cores with 3.25 EC2 Compute Units each), 850 GB of local instance storage, 64-bit platform
- **High-Memory Quadruple Extra Large Instance** 68.4 GiB of memory, 26 EC2 Compute Units (8 virtual cores with 3.25 EC2 Compute Units each), 1690 GB of local instance storage, 64-bit platform
- **High-CPU Instances**
- **High-CPU Medium Instance** 1.7 GiB of memory, 5 EC2 Compute Units (2 virtual cores with 2.5 EC2 Compute Units each), 350 GB of local instance storage, 32-bit or 64-bit platform
- **High-CPU Extra Large Instance** 7 GiB of memory, 20 EC2 Compute Units (8 virtual cores with 2.5 EC2 Compute Units each), 1690 GB of local instance storage, 64-bit platform
- **High Storage Instances**
- **High Storage Eight Extra Large** 117 GiB memory, 35 EC2 Compute Units, 24 * 2 TB of hard disk drive local instance storage, 64-bit platform, 10 Gigabit Ethernet
- **High I/O Instances**
- **High I/O Quadruple Extra Large** 60.5 GiB memory, 35 EC2 Compute Units, 2 * 1024 GB of SSD-based local instance storage, 64-bit platform, 10 Gigabit Ethernet

EC2 Cluster Instances

- **Cluster Compute Instances** provide proportionally high CPU resources with increased network performance and are well suited for High Performance Compute (HPC) applications and other demanding network-bound applications.
 - **Cluster Compute Eight Extra Large** 60.5 GiB memory, 88 EC2 Compute Units, 3370 GB of local instance storage, 64-bit platform, 10 Gigabit Ethernet
- **High Memory Cluster Instances** provide proportionally high CPU and memory resources with increased network performance, and are well suited for memory-intensive applications including in-memory analytics, graph analysis, and scientific computing.
 - **High Memory Cluster Eight Extra Large** 244 GiB memory, 88 EC2 Compute Units, 240 GB of local instance storage, 64-bit platform, 10 Gigabit Ethernet
- **Cluster GPU Instances** provide general-purpose graphics processing units (GPUs) with proportionally high CPU and increased network performance for applications benefitting from highly parallelized processing, including HPC, rendering and media processing applications.
 - **Cluster GPU Quadruple Extra Large** 22 GiB memory, 33.5 EC2 Compute Units, 2 x NVIDIA Tesla “Fermi” M2050 GPUs, 1690 GB of local instance storage, 64-bit platform, 10 Gigabit Ethernet.

EC2 Payment methods

- **On-Demand Instances** let you pay for compute capacity by the hour with no long-term commitments.
 - **Reserved Instances** give you the option to make a low, one-time payment for each instance you want to reserve and in turn receive a significant discount on the hourly charge for that instance.
 - **Spot Instances** allow customers to bid on unused Amazon EC2 capacity and run those instances for as long as their bid exceeds the current Spot Price

AWS Storage Choices



Amazon EFS

File



Amazon EBS

Amazon EC2
Instance Store

Block



Amazon S3

Amazon Glacier

Object

Amazon Elastic Block Store (EBS)

Scalable Block Storage: Provides **block-level storage volumes** ranging from **1 GB to 1 TB** for use with Amazon EC2 instances.

- Multiple volumes can be **attached to the same instance**.

Persistent & Network-Attached

- Volumes are **network-attached** and **persist independently** from the life of an instance.
- Behave like raw, unformatted block devices — you can:
 - Create file systems
 - Use them like traditional hard drives
 - Store application data, databases, or logs

Availability Zone Placement

- EBS volumes are created in a **specific Availability Zone (AZ)**.
- Can only be attached to EC2 instances in the **same AZ**.
-

Amazon Elastic Block Store (EBS)

Built-In Data Protection

- Automatically **replicated within the same AZ** to protect from hardware failures.
- **Point-in-time snapshots** can be created and stored in Amazon S3 for long-term durability.

Snapshots

- Snapshots can:
 - Serve as the **starting point for new volumes**
 - Be **restored multiple times**
 - Be **copied across AWS Regions** for disaster recovery or migration.

EBS Volumes

Standard Volumes (Magnetic Storage)

- Ideal for **applications with moderate or bursty I/O** requirements.
- Deliver **~100 IOPS on average**.
- Well-suited for **boot volumes** — burst capability enables **fast instance start-up**.

Provisioned IOPS (io1 / io2)

- Designed for **predictable, high-performance** storage in **I/O-intensive workloads** (e.g., relational or NoSQL databases).
- You **specify the IOPS rate** during volume creation — guaranteed for the lifetime of the volume.
- Supports **up to 4,000 IOPS per single volume**.
- You can **stripe multiple volumes** together to achieve **tens of thousands of IOPS** per EC2 instance.

EBS Volumes

EBS-Optimized Instances

- To maximize performance, launch EC2 instances as **EBS-Optimized**.
- Provide **dedicated network throughput** between EC2 and EBS.
- Bandwidth options: **500 Mbps to 1,000 Mbps**, depending on instance type.

Pricing Model

- Charged based on:
 - **Per GB-month** of provisioned storage.
 - **Per 1 million I/O requests** (for certain volume types).

Block Storage

Best Suited For

- **Transactional databases** (e.g., MySQL, PostgreSQL)
- **Random read/write workloads**
- **Structured database storage**

Data Organization

- Data is split into **evenly sized blocks** for efficient access.
- **No metadata** is stored within the blocks themselves.
- Maintains only an **index of data blocks** to track their location — does not interpret or “understand” the actual data content.

Object Storage

Data Storage Approach

- Stores files as **complete objects** — **not** divided into blocks.

Object Structure

- Each object contains:
 - **Data** (the actual content)
 - **Metadata** (descriptive information about the data)
 - A **globally unique identifier (GUID)**

Access & Usage

- **Cannot be mounted** like a traditional disk or file system.
- Objects can be **retrieved from anywhere** using their global unique ID, making them ideal for distributed access

Amazon EBS Volume Types

	Solid-State Drives (SSD)		Hard Disk Drives (HDD)	
	General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
Max volume size	16 TiB	16 TiB	16 TiB	16 TiB
Max IOPS/volume	16,000	64,000	500	250
Max throughput/volume	250 MiB/s	1,000 MiB/s	500 MiB/s	250 MiB/s

Amazon EBS Volume Types

Use Cases

Solid-State Drives (SSD)		Hard Disk Drives (HDD)	
General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
• Recommended for most workloads • System boot volumes • Virtual desktops • Low-latency interactive apps • Development and test environments	• Critical business applications that require sustained IOPs performance, or more than 16,000 IOPS or 250 MiB/s of throughput per volume • Large database workloads	• Streaming workloads requiring consistent, fast throughput at a low price • Big data • Data warehouses • Log processing • Cannot be a boot volume	• Throughput-oriented storage for large volumes of data that is infrequently accessed • Scenarios where the lowest storage cost is important • Cannot be a boot volume

Amazon Simple Storage Service (Amazon S3)

- Amazon Simple Storage Service (Amazon S3) is an **object storage service** that offers industry-leading scalability, data availability, security, and performance.
- The **customers of all sizes**, organizations and industries can use it to store and protect any amount of data.
- S3 are used for **variety of uses** like websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics.
- Amazon S3 provides **easy-to-use** management **features**.

Features S3

- **Unlimited number** of objects with **write, read, and delete** facility and containing 1 byte to 5 terabytes of data.
- Each object is stored in a **bucket** and retrieved via a **unique, developer-assigned key**.
- Objects stored in a specific **region never leave** the Region unless transferred.
- **Authentication** mechanisms are provided to ensure that data is kept secure from unauthorized access.
- Uses **standards-based** REST and SOAP interfaces to work with any Internet-development toolkit.
- Built to be **flexible** so that protocol or functional layers can easily be added.
- Provides functionality to simplify manageability of data through its lifetime.

Starting with S3

Launching an Amazon S3 service for the **first time** need the following steps:

- Sign up for AWS
- Create an IAM (Identity and Access Management) user
- Sign in as an IAM user
- Creating a bucket
- Uploading an object to a bucket
- Viewing an object
- Deleting objects and buckets

Create a Bucket

1. **Sign in** to AWS and open the [S3 Console](#).
2. Click **Create bucket**.
3. **Enter a bucket name** (must):
 - Be unique across all of Amazon S3
 - 3–63 characters
 - No uppercase letters
 - Start with a lowercase letter or number
 - Cannot be changed after creation
4. Select the **AWS Region** for the bucket.
5. In **Block Public Access** settings, keep defaults.
6. Click **Create bucket**.

Upload an Object to a Bucket

1. In the **Buckets list**, select the target bucket.
2. On the **Overview** tab, click **Upload** (or **Get Started** if empty).
3. In the **Upload** dialog, choose **Add files**.
4. Select the file from your computer and click **Open**.
5. Click **Upload** to start the transfer.

To Download an Object from a Bucket

In the **Buckets list**, select the bucket.

In the **Name list**, click the object you want to download.

The **Object Overview** panel will open.

On the **Overview** tab:

- **Open** → View the object in your browser.
- **Download** → Save the object to your computer.

Delete an object and Bucket

Delete an Object

1. Open the **Buckets list** and select the bucket containing the object.
2. In the **Name list**, check the box for the object to delete.
3. Choose **Actions** → **Delete**.
4. Confirm the object name in the dialog box, then click **Delete**.

Delete a Bucket

1. In the **Buckets list**, select the bucket.
2. Choose **Delete** and confirm.

Amazon Simple Storage Service (S3)

- Simple **web services interface** to store and retrieve **any amount of data, anytime, anywhere**.
- Supports **objects from 1 byte to 5 TB**; unlimited total objects.
- Objects are stored in **buckets** and identified by a **unique key**.

Bucket & Region

- Choose from multiple **AWS Regions** to optimize **latency, cost, or compliance**.
- Objects remain in their Region unless explicitly transferred.

Security & Access Control

- Built-in **authentication mechanisms**.
- Objects can be **private, public**, or shared with **specific users**.

Pricing

- Charged based on:
 - **Per GB-month storage**
 - **Per I/O request**
 - **Per data modification request**

Securing Amazon S3 buckets and objects

- Newly created S3 buckets and objects are **private** and **protected** by default
- When use cases must share Amazon S3 data –
 - Manage and control the data access
 - Follow the **principle of least privilege**
- Tools and options for controlling access to Amazon S3 data –
 - [Block Public Access](#) feature: It is enabled on new buckets by default, simple to manage
 - [IAM policies](#): A good option when the user can authenticate using IAM
 - [Bucket policies](#): You can define access to a specific object or bucket
 - [Access control lists](#) (ACLs): A legacy access control mechanism
 - [S3 Access Points](#): You can configure access with names and permissions specific to each application
 - [Presigned URLs](#): You can grant time-limited access to others with temporary URLs
 - [AWS Trusted Advisor](#) bucket permission check: A free feature



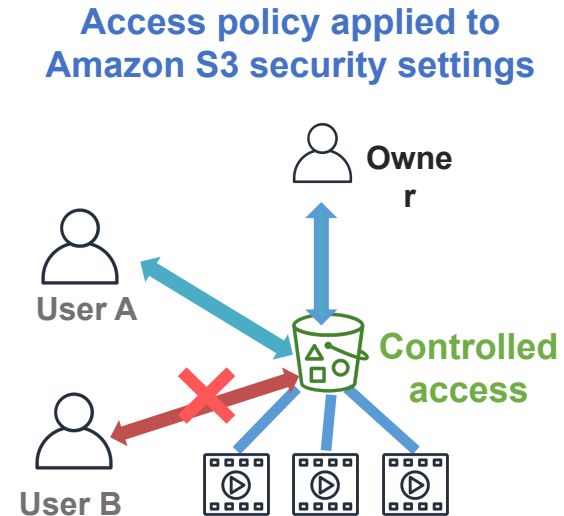
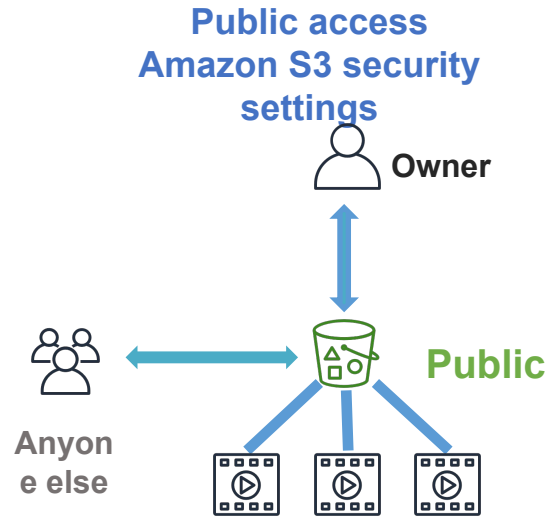
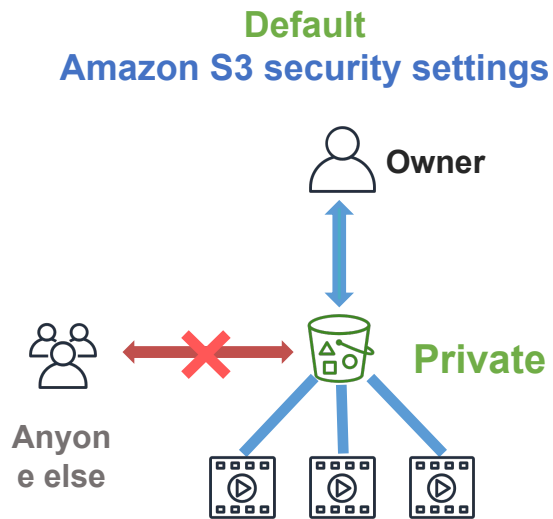
Controlling Access to Amazon S3

- **Default Privacy:** All buckets are private by default; explicit access is required.
- **Block Public Access:** A safeguard that overrides all other permissions; enable it for buckets that should never be public.
- **IAM Policies:** Identity-based rules that grant access to specific users or roles.
- **Bucket Policies:** Resource-based rules for granting access (cross-account or anonymous). Must be written and tested carefully; can also enforce explicit denies.
- **Access Points:** Provide custom hostnames with unique permissions, useful for shared datasets and multi-application access.
- **ACLs (Access Control Lists):** Older method; use sparingly and avoid overly permissive settings.
- **AWS Trusted Advisor:** Helps identify buckets that might have unsafe global access.

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Three General Approaches to Configuring Access

Configure the appropriate security settings for your use case on the bucket and objects.



Consider Encrypting Objects in Amazon S3

- **Encryption** encodes data with a **secret key**, which makes it unreadable
 - Only users who have the secret key can decode the data
 - Optionally, use AWS Key Management Service (AWS KMS) to manage secret keys
- *Server-side* encryption
 - On the bucket, enable this feature by selecting the Default encryption option
 - Amazon S3 encrypts objects before it saves the objects to disk, and decrypts the objects when you download them
- *Client-side* encryption
 - Encrypt data on the client side and upload the encrypted data to Amazon S3
 - In this case, you manage the encryption process



Data Encryption in Amazon S3

- **Purpose:** Protects digital data by converting it into unreadable form using a secret key. Without the key, the data is meaningless — even if stolen.

Encryption Options in S3

1. Server-Side Encryption (SSE)

- Set via **Default Encryption** on a bucket.
- S3 encrypts data **before storing** and decrypts it **on retrieval**.

2. Client-Side Encryption (CSE)

- Data is encrypted **before upload** by the client.
- You manage the encryption keys and process.

Both methods ensure data remains protected by keeping encryption keys separate from stored data.

Amazon S3 benefits



Durability

- It ensures data is not lost
- S3 Standard storage provides 11 9s (or 99.999999999%) of durability



Scalability

- It offers virtually unlimited capacity
- Any single object of 5 TB or less



Availability

- You can access your data when needed
- S3 Standard storage class is designed for four 9s (or 99.99%) availability



Security

- It offers fine-grained access control



Performance

- It is supported by many design patterns

Key Features of Amazon S3

Durability

- **11 nines (99.999999999%)** – near-zero chance of object loss.
- Data is redundantly stored across **multiple devices & facilities**.
- Automatic detection & repair of lost redundancy with **checksums**.

Availability

- **99.99% uptime** for data access.
- Scales to **virtually unlimited storage capacity**.

Security

- Fine-grained **access controls** and **encryption** options.

Performance

- **Low latency** (milliseconds to first byte) for most storage classes.
- Supports caching, retry logic, and request parallelization for high throughput.

Amazon Simple Queue Service (SQS)

- Amazon Simple Queue Service (SQS) is a **message queuing service**.
- It enables you to **decouple and scale** microservices, distributed systems, and server less applications.
- SQS **eliminates** the complexity and overhead associated with managing and operating message oriented middleware.
- Using SQS, you can **send, store, and receive messages** between software components at any volume.
- SQS offers two types of message queues.
 - *Standard queues* offer maximum throughput, best-effort ordering, and at-least-once delivery.
 - *SQS FIFO* queues are designed to guarantee that messages are processed exactly once, in the exact order that they are sent.

- **Unlimited** queues and messages.
- **Payload Size:** Message payloads can contain up to **256KB** of text in any format.
- **Batches:** Send, receive, or delete messages **in batches** of up to 10 messages or 256KB.
- Batches cost the same amount as single messages, meaning SQS can be even **more cost effective** for customers that use batching.
- **Long polling:** Reduce **extraneous polling** to minimize cost while receiving new messages as quickly as possible.
- Retain messages in queues for up to **14 days**.
- Send and receive messages **simultaneously**.

- **Message locking:** When a message is received, it becomes “locked” while being processed. This keeps other computers from processing the message simultaneously.
- **Queue sharing:** Securely share Amazon SQS queues anonymously or with specific AWS accounts.
- **Server-side encryption (SSE):** Protect the contents of messages in Amazon SQS queues using keys managed in the AWS Key Management Service (AWS KMS). SSE encrypts messages as soon as Amazon SQS receives them. The messages are stored in encrypted form and Amazon SQS decrypts messages only when they are sent to an authorized consumer.
- **Dead Letter Queues (DLQ):** Handle messages that have not been successfully processed by a consumer with Dead Letter Queues.

Launching an Amazon SQS service for the first time need the following steps:

- Sign up for AWS

- Create an IAM user

- Get your access key ID and secret access key

- Create a Queue

- Send a Message

- Receive a Message

- Delete a Message

- Delete a Queue

To create an AWS account

- Open <https://portal.aws.amazon.com/billing/signup>.

- Follow the online instructions.

Create an IAM user with administrative privilege.

Sign in to the Amazon SQS console.

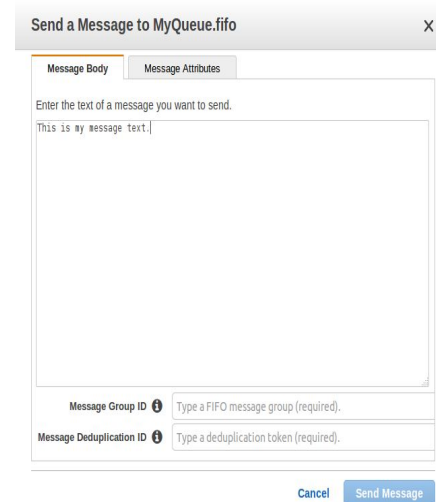
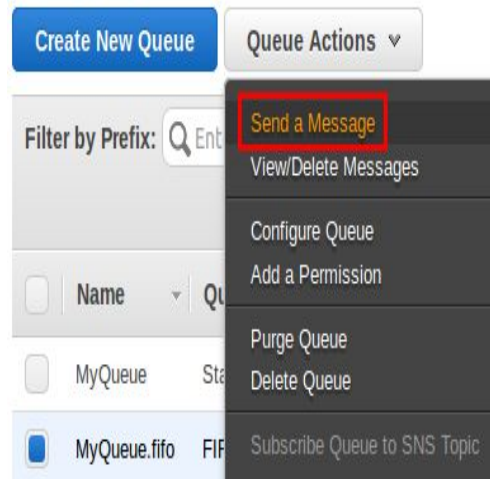
- Choose **Create New Queue**.
- On the **Create New Queue** page, ensure that you're in the correct region and then type the **Queue Name**. The name of a FIFO queue must end with the *.fifo* suffix.
- **Standard** is selected by default.
- To create your queue with the default parameters, choose **Quick-Create Queue**.
- Your new queue is created and selected in the queue list.
- The **Queue Type** column helps you distinguish standard queues from FIFO queues at a glance.
- Your queue's **Name**, **URL**, and **ARN** are displayed on the **Details** tab.

Send a Message

From the queue list, select the queue that you've created.

From Queue Actions, select Send a Message.

The **Send a Message** to QueueName dialog box is displayed.



View/Delete Messages

- From the queue list, **select** the queue that you have created.
- From **Queue Actions**, select View/Delete Messages.
- Choose **Start Polling** for messages.
- Amazon SQS begins to poll the messages in the queue. The dialog box displays a message from the queue. A progress bar at the bottom of the dialog box displays the status of the message's visibility timeout.
- When the progress bar is filled the message becomes visible to consumers.
- **Before the visibility timeout expires**, select the message that you want to delete and then choose Delete 1 Message.
- In the **Delete Messages** dialog box, confirm that the message you want to delete is checked and choose Yes, Delete Checked Messages.
- The selected message is deleted.
- Select Close.

Create New Queue

Queue Actions

Filter by Prefix:

☐ Name

Queue

☐ MyQueue

State

☒ MyQueue.fifo

FIFO

Send a Message

View/Delete Messages

Configure Queue

Add a Permission

Purge Queue

Delete Queue

Subscribe Queue to SNS Topic

View/Delete Messages in MyQueue.fifo

View up to: messages

Poll queue for: seconds

Start Polling for Messages

Stop Now

Polling for new messages once every 2 seconds.

View/Delete Messages in MyQueue.fifo

View up to: messages

Poll queue for: seconds

Start Polling for Messages

Stop Now

Polling for new messages once every 2 seconds.

Delete	Body	Message Group ID	Message Deduplication ID	Sequence Number
☒	This is my message text.	MyMessageGroupI...	MyMessageDeduplicati...	188259353925

54%

Polling the queue at 0.6 receives/second. Stopping in 13.7 seconds. Messages shown above are currently hidden from other

Close

Delete 1 Message

View/Delete Messages in MyQueue.fifo

View up to: messages

Poll queue for: seconds

Polling for Messages...

Stop Now

Polling for new messages once every 2 seconds.

Delete	Body	Message Group ID	Message Deduplication ID	Sequence Number
☐	This is my message text.	MyMessageGroupI...	MyMessageDeduplicati...	188259353925

54%

Polling the queue at 0.6 receives/second. Stopping in 13.7 seconds. Messages shown above are currently hidden from other

Close

Delete Messages

Delete Messages

Are you sure you want to delete the following message?

You may uncheck messages that you do not want to delete.

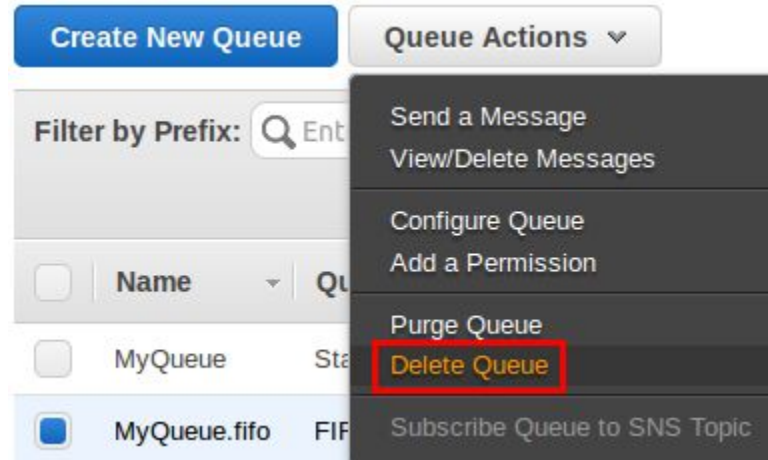
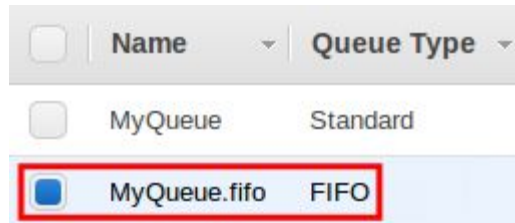
☒ This is my message text. (24 bytes)

Cancel

Yes, Delete Checked Messages

Delete Queue.

From the queue list, select the queue that you have created.
From Queue Actions, select Delete Queue.



Google AppEngine

- Google App Engine (GAE) is a cloud-based platform that allows developers to build, deploy, and manage web applications using Google's infrastructure.

Some of the Key Features of Google App Engine

- **Scalability:** GAE provides automatic scaling, which means that it can automatically adjust the resources used by an application in response to changes in traffic.
- **Reliability:** GAE is built on top of Google's infrastructure, which has a proven track record of reliability. This infrastructure is designed to be fault-tolerant, meaning that if one component fails, the system can automatically switch to a backup component.
- **Flexibility:** GAE supports multiple programming languages, including Java, Python, Go, and PHP. GAE also supports a variety of development frameworks, such as Django, Flask, and Spring, making it easier for developers to build and deploy their applications.
- **Other Features:** GAE provides built-in services such as NoSQL datastore, Memcache, task queues, and cron jobs, which allow developers to quickly and easily add functionality to their applications without worrying about managing the underlying infrastructure.

Developing Applications on Google App Engine Involves the Following Steps

- **Choose a programming language and framework:**
 - Google App Engine supports several programming languages, including Java, Python, Go, and Node.js. You can choose a language and framework based on your expertise and the requirements of your application.
- **Set up your development environment:**
 - To develop applications on App Engine, you need to set up your development environment with the appropriate SDKs, tools, and libraries. Google provides SDKs for each supported language that you can download and install on your local machine.
- **Create a new App Engine project:**
 - To create a new App Engine project, you need to sign up for a Google Cloud Platform account and enable the App Engine service. You can then create a new project and configure its settings, such as the project ID, region, and billing account.
- **Write your application code:**
 - You can write your application code using your chosen programming language and framework. App Engine provides APIs and libraries that you can use to interact with the App Engine environment, such as the datastore API for accessing the NoSQL database.

Developing Applications on Google App Engine Involves the Following Steps

- **Test your application locally:**
 - You can test your application locally by running it in a local development server provided by the App Engine SDK. This allows you to test your application's functionality and performance before deploying it to the App Engine environment.
- **Deploy your application:**
 - To deploy your application to the App Engine environment, you need to use the App Engine SDK or the Google Cloud Console. You can specify the version, traffic splitting, and other deployment settings. Once deployed, your application is accessible to users via a unique URL.
- **Monitor and manage your application:**
 - You can monitor and manage your application using the App Engine console or the Cloud Console. This allows you to view logs, metrics, and errors, and make configuration changes or updates as needed.

Windows Azure

- Windows Azure is a cloud computing platform developed by Microsoft for building, deploying, and managing applications and services through a global network of data centers.
- It provides a range of cloud-based services, IaaS, PaaS, and SaaS, making it a flexible and scalable option for organizations of all sizes.
- It supports multiple programming languages and development tools, including .NET, Java, PHP, Node.js, and Python, allowing developers to use their preferred tools and frameworks.
- With Windows Azure, organizations can quickly and easily create, deploy, and manage applications and services, while reducing the costs and complexities of traditional on-premises infrastructure.

For more details: <https://www.geeksforgeeks.org/microsoft-azure/>

Comparison of Cloud Computing Platforms

Feature	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform
Compute Services	EC2, Lambda, Batch	VMs, Functions	Compute Engine, Cloud Functions
Storage Services	S3, EBS, Glacier	Blob, File, Disk, Archive	Cloud Storage, Cloud SQL, Cloud Spanner
Database Services	RDS, DynamoDB, Redshift	SQL Database, Cosmos DB	Cloud SQL, Cloud Datastore, Cloud Bigtable
Networking Services	VPC, Direct Connect, Route 53	Virtual Network, ExpressRoute, Load Balancer	Virtual Private Cloud, Cloud Load Balancing
ComAnalytics and Big Data Services	EMR, Kinesis, Redshift	HDInsight, Stream Analytics	BigQuery, Dataflow, Pub/Sub

Comparison of Cloud Computing Platforms

Feature	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform
Machine Learning Services	SageMaker, Rekognition, Comprehend	Azure ML, Cognitive Services	Cloud ML Engine, Cloud AutoML
Internet of Things Services	IoT Core, Greengrass	IoT Hub, IoT Edge	Cloud IoT Core, Cloud Pub/Sub
DevOps and Management Services	CloudFormation, CodeDeploy	Azure DevOps, Azure Resource Manager	Cloud Deployment Manager, Stackdriver Monitoring

based on features available as of September 2021

Table for the Comparison of Cloud Computing Platforms size and performance wise

Feature	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform
Market Share	Largest	Second-largest	Third-largest
Global Infrastructure	Massive	Rapidly expanding	Rapidly expanding
Hybrid Cloud Solutions	Limited	Strong focus	Limited
Big Data and Machine Learning	Strong offerings	Strong offerings	Expertise in these areas
Kubernetes Support	Strong offerings	Strong offerings	Strong offerings
Fiber Optic Cable Investments	Significant investments	N/A	Significant investments
Pricing	Complex pricing model	Complex pricing model	Simplified pricing model
Customer Support	Industry-leading support	Strong support	Strong support

<https://docs.aws.amazon.com/index.html>

<https://aws.amazon.com/free/>